Lesson 10.4 - Test Statistic for Means and Population Proportions

> **Video:** Test Statistic For Means and Population Proportions by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

The big picture: turning a sample into one number

In Lesson 10.3 we drew the line: the **critical value** (like ± 1.96) that separates "ordinary" from "surprising". This lesson answers the other half: **where does our sample land on that same scale?**

1. The problem: different units. Say H_0 : $\mu = 100$ g and our sample of chip bags averages $\bar{x} = 98$ g. We can't compare "98 grams" with "1.96" directly. They're on different scales. We need to convert the sample into a z-score (or t-score), the same scale as the critical value.

2. The idea: count standard errors. The **test statistic** answers one question: *how many standard errors is the sample away from what H_0 claims?*

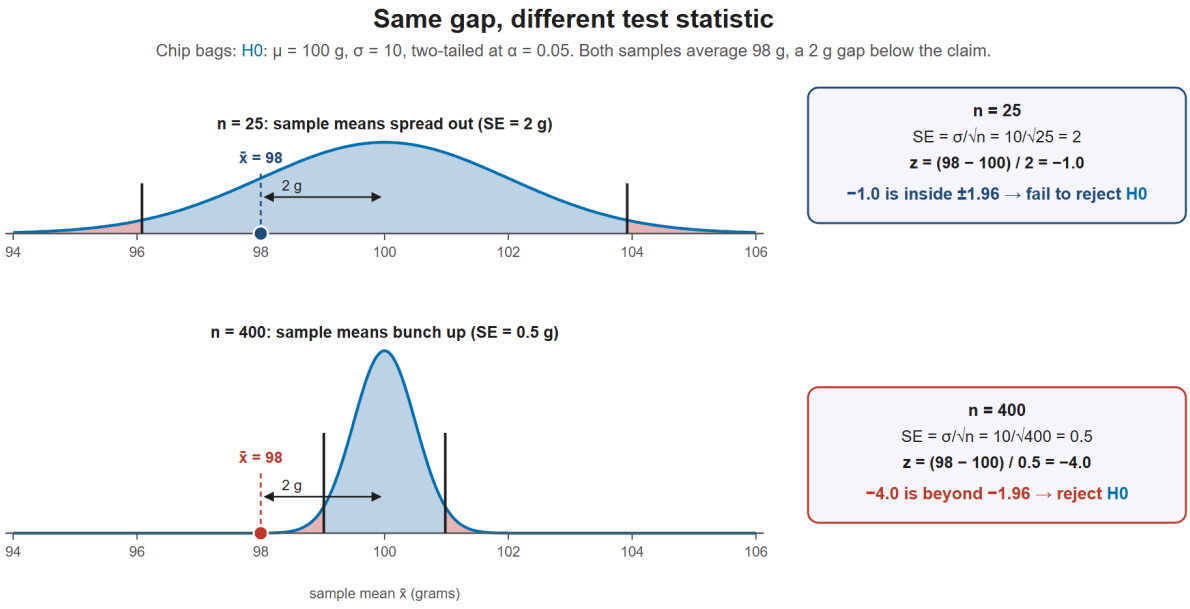
$$\text{test statistic} = \frac{\text{what we got} - \text{what } H_0 \text{ claims}}{\text{standard error}}$$

- The **top** is the gap: how far the sample is from the claim ($98 - 100 = -2$ g).
- The **bottom** is the **standard error (SE)**: how much sample results normally wobble by chance alone. For a mean, $SE = \sigma/\sqrt{n}$.

3. Why divide by the standard error? Because a 2 g gap can be ordinary or shocking, depending on how much sample means normally wobble.

- With $n = 25$ bags ($\sigma = 10$): $SE = \sigma/\sqrt{n} = 10/\sqrt{25} = 2$, so $z = \dfrac{\bar{x} - \mu}{SE} = -2/2 = -1.0$. That's a normal wobble → fail to reject.
- With $n = 400$ bags: $SE = \sigma/\sqrt{n} = 10/\sqrt{400} = 0.5$, so $z = \dfrac{\bar{x} - \mu}{SE} = -2/0.5 = -4.0$. That's way out in the tail → reject.

Same gap, different verdict. Bigger samples wobble less ($\sqrt{n}$ is in the denominator), so the same gap becomes more convincing.



Top: with 25 bags, sample means spread widely around 100, so 98 is well inside the fail-to-reject region. Bottom: with 400 bags, sample means bunch tightly around 100, so 98 is far beyond the critical value. The gap is 2 g in both cases.

4. Why use H_0's value in the formula? The test pretends H_0 is true (Lesson 10.3), so we measure the distance **from H_0's claim**. That's why μ (or p) in every formula below is the hypothesized value, not something from the sample.

5. z or t? If you know the population σ, the statistic follows the normal curve: use **z**. If you have to estimate it with the sample's s, that estimate adds extra uncertainty, especially for small samples. The **t-distribution** accounts for this. It has fatter tails, so its critical values are a bit larger than z's.

Where this lesson fits in a full test:

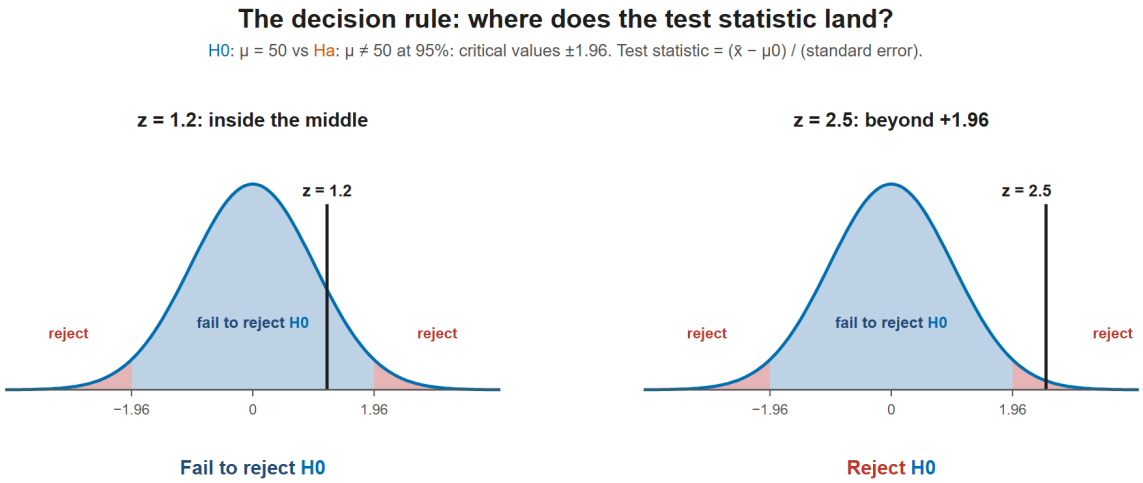
1. Write H_0 and H_a (Lesson 10.1).
2. Choose α.
3. Use the sign in H_a to find the critical value(s) (Lesson 10.3).
4. **Compute the test statistic from the sample.** ← this lesson
5. Compare: beyond the critical value → reject H_0. Otherwise → fail to reject H_0.

> **Same building blocks as confidence intervals (Chapter 9).** A confidence interval is *estimate* ± *critical value* × *SE*. A test statistic is *(estimate − claim)* ÷ *SE*. Same three pieces, rearranged.

The decision rule

1. Find the **critical value(s)** from the significance level and the type of test. For a two-tailed test at 95%, ± 1.96.
2. Compute the **test statistic** z_c (or t) from the sample.
3. Statistic in the rejection region → **reject** H_0. Statistic in the fail-to-reject region → **fail to reject** H_0.

Example: $H_0 : \mu = 50$ vs $H_a : \mu \neq 50$ at 95% has critical values ± 1.96. A calculated z_c beyond 1.96 (or below −1.96) means reject.



Compute the test statistic, then see which region it lands in. Between −1.96 and 1.96, fail to reject H_0. Beyond either critical value, reject H_0.

Test statistic for a mean

Sample size	σ known?	Distribution	Test statistic
$n < 30$	**No** (only s)	**t**, df $= n - 1$	$t = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$
$n < 30$	Yes	z	$z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
$n \geq 30$	No (use s)	z (approximately normal)	$z = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$
$n \geq 30$	Yes	z	$z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

Here μ is the value claimed in H_0.

> Many textbooks use the t-distribution **whenever σ is unknown**, whatever the sample size. For $n \geq 30$, t and z give nearly the same answer, so the video's rule is a common shortcut.

Test statistic for a proportion

$$z = \frac{\hat{p} - p}{\sqrt{\dfrac{pq}{n}}}, \qquad q = 1 - p$$

- $\hat{p}$ = sample proportion (x/n)
- p = the proportion claimed in H_0 (sometimes written p_0). Use this hypothesized value under the square root, not $\hat{p}$.

Key takeaways

- Test statistic = (sample value − hypothesized value) ÷ standard error = "how many standard errors from H_0".
- The same gap is more convincing with a bigger sample, because the standard error shrinks like $1/\sqrt{n}$.
- Mean: use t with s when $n < 30$ and σ is unknown. Otherwise use z.
- Proportion: $z = (\hat{p} - p)/\sqrt{pq/n}$, with p from H_0.
- Compare with the critical value(s) to reject or fail to reject H_0.